

Software Architecture & Design

Topics covered

1. Introduction to Software Architecture
2. Architectural Structures and Views
3. Architecture Business Cycle (ABC)

Question 1 (10 marks)

Define: a) Software Architecture

b) Software Design

c) Software Architect

d) Quality Attribute

e) Architectural Pattern

Question 2 (10 marks)

Differentiate Software Architecture and Software Design.

Differentiate Structure and View.

Question 3 (10 marks)

Explain five quality attributes with examples.

Question 4 (10 marks)

List five stakeholders and explain their concerns.

Question 5 (15 marks)

Explain Module Structure.

Explain Component-and-Connector Structure.

Explain Allocation Structure.

Question 6 (15 marks)

Explain Decomposition Structure with an example.

Explain Layered Structure with an example.

Explain Client-Server Structure with an example.

Question 7 (15 marks)

Explain Architecture Business Cycle (ABC) and draw the diagram.

Question 8 (15 marks)

Discuss how stakeholder requirements, developing organization and technical environment influence a university Learning Management System architecture.